

ONLINE EXAMINATION SYSTEM

Technical Documentation and Implementation Guide

1. System Overview

The Online Examination System (OES) is a console-based C++ application designed to conduct digital exams efficiently. It provides a structured environment for educational assessment through a command-line interface.

Key Capabilities:

- Student authentication and exam attempts
- Multiple Choice Question (MCQ) based testing
- Automatic result calculation and grading
- Permanent storage for performance tracking
- Administrative control for question bank management

2. Core Concepts Used

- **Object-Oriented Programming (OOP):** Use of encapsulation to define Student, Exam, Question, and Result entities.
- **Static Members:** Tracking global system data such as total registered students and exams conducted.
- **File Handling:** Persistent storage for the question bank and student results using `fstream`.
- **Composition:** The Exam class integrates multiple Question objects to manage test instances.
- **Arrays of Objects:** Management of multiple data entries for users and questions.
- **Control Structures:** Implementation of a menu-driven navigation system using switch statements.

3. System Architecture

```
Online Exam System
|
|— Student Class
|   |— Login
|   |— Attempt Exam
|   |— View Result
|
|— Question Class
|   |— Question Text
|   |— Options
|   |— Correct Answer
|
|— Exam Class (Composition)
|   |— Manages Questions
|   |— Conducts Exam Flow
|
|— Admin Class
|— Add Questions
|— Manage Exams
```

4. Class Descriptions

Question Class

Responsible for individual MCQ data management.

- Attributes: Question text, Options (A, B, C, D), Correct Answer.
- Functions: Display question, validate user input.

Exam Class

The central controller for test execution utilizing composition.

- Features: Aggregates Question objects, calculates final scores, manages the logical flow of the session.

Student Class

Handles the user-facing interface for candidates.

- Functions: Session login, exam attempt logic, result retrieval.

Admin Class

Provides administrative overhead and system configuration.

- Functions: Data entry for questions, question bank maintenance, system statistics tracking.

5. C++ Source Code Implementation

```
#include <iostream>
#include <vector>
#include <string>
#include <fstream>

using namespace std;

class Question {
public:
    string qText;
    string options[4];
    char correctOption;

    void setQuestion(string q, string o1, string o2, string o3, string o4, char ans)
    {
        qText = q;
        options[0] = o1; options[1] = o2; options[2] = o3; options[3] = o4;
        correctOption = ans;
    }

    bool checkAnswer(char userAns) {
        return (toupper(userAns) == toupper(correctOption));
    }

    void display() {
        cout << "
" << qText << endl;
        for(int i=0; i<4; i++) {
            cout << (char)('A'+i) << ". " << options[i] << endl;
        }
    }
};
```

```

    }
};

class Exam {
private:
    vector<Question> questions;
    int score;
public:
    Exam() : score(0) {}

    void addQuestion(Question q) {
        questions.push_back(q);
    }

    void start() {
        char choice;
        for(int i=0; i < questions.size(); i++) {
            questions[i].display();
            cout << "Your Answer: ";
            cin >> choice;
            if(questions[i].checkAnswer(choice)) {
                score++;
            }
        }
        cout << "
Exam Finished. Your Score: " << score << "/" << questions.size() << endl;
    }
};

int main() {
    cout << "--- Online Examination System ---" << endl;
    Exam midterm;

    Question q1;
    q1.setQuestion("What is C++?", "Language", "Fruit", "Car", "City", 'A');
    midterm.addQuestion(q1);

    int mode;
    cout << "1. Student Login
2. Admin Exit
Enter choice: ";
    cin >> mode;

    if(mode == 1) {
        midterm.start();
    }

    return 0;
}

```

```
}
```